SAMPATH THOTA

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SUMMARY

Cybersecurity Engineer with 4+ years of experience in **SOC detection engineering**, **cloud security**, and **DevSecOps** across finance enterprise and platform environments. Built **end-to-end security workflows** covering **SIEM-based threat detection**, **automated incident response**, and **cloud-native application security** using **AWS** and **Splunk**. Applied **MITRE ATT&CK**, **Zero Trust**, and **infrastructure-as-code** practices to reduce **detection gaps**, eliminate **false positives**, and improve **system resilience**. Bridges **security operations** and **engineering** by delivering **scalable**, **automation-driven solutions** that improve **response time** and reduce **risk** in **production environments**.

PROFESSIONAL EXPERIENCE

Cyber Security Engineer | Morgan Stanley | USA — Sep 2025 – Present

- Optimized **Splunk** and **Microsoft Sentinel** detection workflows for high-volume financial systems, ingesting **AWS CloudTrail** and EDR telemetry to streamline **incident triage**, contributing to a reduction in response time from **5 to 3.2 hours**.
- Developed and tuned **MITRE ATT&CK-aligned correlation rules** and risk-based alerts for financial threats, reducing 120+ recurring false positives monthly and improving Tier 1 triage efficiency in 24/7 regulated environments.
- Constructed **SOAR-based incident response playbooks** (Python + REST APIs), enabling EC2 isolation and IAM credential revocation within 15 minutes for high-severity incidents affecting critical applications and data.
- Supported **AWS cloud security posture improvements** by enforcing least-privilege IAM and CloudTrail controls, remediating 110+ misconfigurations across S3, RDS, and KMS aligned with SOX and PCI-DSS standards.

Cyber Security Engineer | Airbnb | India — Aug 2021 – Dec 2023

- Implemented **Zero Trust security architecture** across **AWS-based platform** services using **Terraform** and **IAM policies**, enforcing least-privilege access and reducing unauthorized access attempts across internal APIs.
- Engineered a **DevSecOps pipeline** integrating Burp Suite and OWASP ZAP into CI/CD workflows, reducing vulnerability remediation SLA from 30 to 7 days and lowering false-positive findings by 70% in automated scans.
- Secured containerized microservices by deploying **runtime protection controls and WAF rules** across 150+ public-facing endpoints, mitigating critical **OWASP** Top 10 vulnerabilities and strengthening application-layer defenses.
- Consolidated **multi-cloud secrets management** by standardizing vault architecture and access controls, reducing 25 fragmented secret stores to 6 centralized systems and minimizing credential exposure risks.
- Streamlined **compliance reporting (GDPR/CCPA)** using Python and Ansible, generating **16+ recurring audit reports** and reducing manual reporting effort by 200+ hours annually.
- Directed proactive **threat hunting** missions and **red/blue team exercises** to strengthen detection capabilities, integrating **SOAR workflows** that enabled the team to remediate **25+ critical vulnerabilities** and reduce dwell time.

Junior Cyber Security Engineer | Dell Technologies | India — Jan 2020 – Jul 2021

- Implemented **network segmentation and access controls** using Active Directory Group Policies and Dell PowerEdge infrastructure, reducing lateral movement risk across 850+ enterprise endpoints.
- Supported deployment of **PowerProtect Cyber Recovery vaults** using **AWS CloudFormation** and **Terraform**, enabling immutable backup isolation across 5 data centers and strengthening ransomware resilience for 12TB of critical data.
- Developed **Python-based SIEM integrations** with **CrowdStrike Falcon**, creating **60+ custom indicators** of compromise (IOCs) for **threat hunting** and automating **incident triage** to reduce endpoint response time from 4.2 hours to 45 minutes.
- Generated **secure development lifecycle (SDL) security testing** using Ansible and Jenkins pipelines, integrating Metasploit, Burp Suite, and OWASP ZAP across 25+ firmware releases and improving vulnerability detection coverage.
- Hardened enterprise infrastructure by applying **STIG-based security configurations** via PowerShell and Active Directory across 150+ servers, and supported incident response investigations using CyberSense analytics for 45+ security events.

TECHNICAL SKILLS

- **Security Monitoring & Detection:** Splunk Enterprise SIEM, Microsoft Sentinel, Log Analysis, Alert Triage, Incident Response (NIST 800-61), MITRE ATT&CK Mapping, Threat Intelligence Integration, SOAR Playbooks, EDR/XDR, Secure Telemetry
- **Cloud & Infrastructure Security:** AWS (IAM, EC2, S3, RDS, CloudTrail, KMS), Azure AD, RBAC, Cloud Security Posture Management (CSPM), Terraform Security Controls, Infrastructure as Code (IaC) Review
- **Risk & Compliance Frameworks:** NIST Cybersecurity Framework (CSF), NIST 800-53, ISO/IEC 27001, CIS Critical Security Controls, SOC 2 Control Testing, SOX ITGC Support, Audit Evidence Management
- **Vulnerability & Application Security:** Nessus, Qualys, Burp Suite, OWASP ZAP, Patch Management, Secure Configuration Baselines, Web Application Security Testing
- **Identity & Endpoint Security:** Identity & Access Management (IAM), Active Directory, Group Policy, Data Loss Prevention (DLP), Encryption at Rest and in Transit, Access Reviews
- **Automation & DevSecOps:** Python, Bash, Jenkins, GitHub Actions, AWS CodePipeline, CI/CD Security Controls, Secure API Integration
- **Networking & Systems:** Linux Security Hardening, Firewall Configuration, TCP/IP, DNS Security, Wireshark Packet Analysis, TLS/SSH, Access Control Lists (ACLs), Nmap, OpenVAS

EDUCATION

Master of Science in Cyber Security | University of Alabama at Birmingham | Birmingham, Alabama — Jan 2024 – May 2025

CERTIFICATIONS

- **ISC2 – Certified in Cybersecurity (CC)**
- **Qualys – Vulnerability Management**
- **Google – Google Cybersecurity**
- **CompTIA Security+ (In Progress)**